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COSKINOLINA SISTANENSIS N. SP., A NEW LARGER BENTHIC FORAMINIFERA FROM THE EARLY EOCENE OF EASTERN IRAN

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Abstract A new conical agglutinating foraminifera is described as *Coskinolina sistanensis* n. sp. from Eocene shallow-water carbonates of the Birjand region, Sistan Suture Zone of Eastern Iran. The dimorphic tests of the new taxon are characterized by a comparably small initial spire in both generations, a small megalosphere in the A-form, a regular structure of the loosely ordered straight pillars, and the low number of comparably high chambers. Within the studied section, *C. sistanensis* n. sp. is restricted to the middle Ilerdian to late Cuisian based on alveolinids, nummulitids, and orthophragminids (SBZ 8–12).

Keywords: Sistan Suture Zone, Taxonomy, Coskinolinidae, biostratigraphy

INTRODUCTION

Larger benthic foraminifera (LBF) such as alveolinids, nummulitids, orthophragminids, and conical agglutinating taxa are of particular importance for dating Paleogene shallow-water carbonates (SBZ 1-20; Serra-Kiel et al., 1998). The most comprehensive research on LBF of Eocene carbonates of Iran were conducted by Rahaghi (1978, 1980, 1983), Rahaghi and Schaub (1976) and Hottinger (2007: Jahrum Formation, Zagros Zone). Recent studies on LBF from Ilerdian-Cuisian sedimentary successions delivered new data especially on nummulitids, and alveolinids (Hadi et al., 2015; Hadi et al., 2016; Mosaddegh et al., 2017). These investigations yielded also a new conical agglutinating taxon that is described in the present paper as *Coskinolina sistanensis* n. sp.

GEOLOGICAL SETTING

Iran has been divided into several tectono-stratigraphic units, each of which characterized by a relatively unique record of stratigraphy, magmatic activities, metamorphism, orogenic events, tectonics, and overall geological style (Eftekharnezhad, 1980; Alavi 1993, Aghanabati, 2004). In a more simplified separation, eastern Iran can be divided into two parts, the Lut Block and Flysch or the colored mélange of Zabol-Baluch zone. It is also called East Iranian Ranges (Berberian, 1977, Alavi, 1991), Flysch zone (Eftekharnezhad, 1980; Ghorbani, 2013), Nehbandan-Khash zone, Zabol-Balouch zone (Berbarian and King, 1981), Iranshahr-Birjand region (Reyre and Mohafez, 1972) and Sistan Suture Zone (Tirrul et al., 1983; Walker et al., 2009). The Sistan Suture Zone extends as a N-S trending belt over more than 700 km along the border area between the Lut (Iran) and Afghan blocks (Afghanistan), which is as result of eastward-directed subduction of a Neotethyan ocean basin beneath the latter (Bröcker et al., 2013). The lithostratigraphic units in the Sistan ocean zone are substantially composed of Cretaceous ophiolitites and ophiolitic mélanges, overlain by Maastrichtian-Eocene flysch sediments (Babazadeh and De Wever, 2004; Fotoohi-Rad et al., 2009). The studied section containing the new taxon is here called Chenesht section, situated at the northeastern margin of the Lut block, southeast of Birjand city (Fig.1). The section is located about 1 km northwest of Chenesht village, and about 50 km southeast of Birjand city. It is placed in sheet-7855 (coordinates: 32° 38' 22" latitude; 59° 23' 54" longitude) and consists of about 30 m of grey- and brown-colored limestones with distinct beds often less than 60 cm in thickness (Fig. 2). The base is partly covered or consists of volcanoclastic sediments. The Eocene carbonates are overlain by sandstones. The Chenesht section comprises the middle Illerdian (SBZ 8) to late Cuisian (SBZ 12) stratigraphic interval (Fig. 2). The Eocene carbonates of the Sistan Suture Zone have so far not been lithostratigraphically defined (formation, type-locality, etc.).

MATERIAL

The specimens illustrated in this paper comprise 4 thin-sections of the Chenesht section. They are hosted by the Department of Geology, Faculty of Science, Ferdowsi University of Mashhad, under the depository numbers CH 1 to CH 4.

SYSTEMATICS

The high-rank classification follows Pawlowski et al. (2013). For the low-rank classification see Kaminski (2014).

Phylum Foraminifera d' Orbigny, 1826
Class Globothalamea Pawlowski et al., 2013
Order Loftusiida Kaminski & Mikhalevich, 2004
Suborder Orbitolinina Kaminski, 2004
Superfamily Coskinolinoidea Moullade, 1965
Family Coskinolinidae Moullade, 1965
Genus *Coskinolina* Stache, 1875
Type-species *Coskinolina liburnica* Stache, 1875
Coskinolina sistanensis n. sp.

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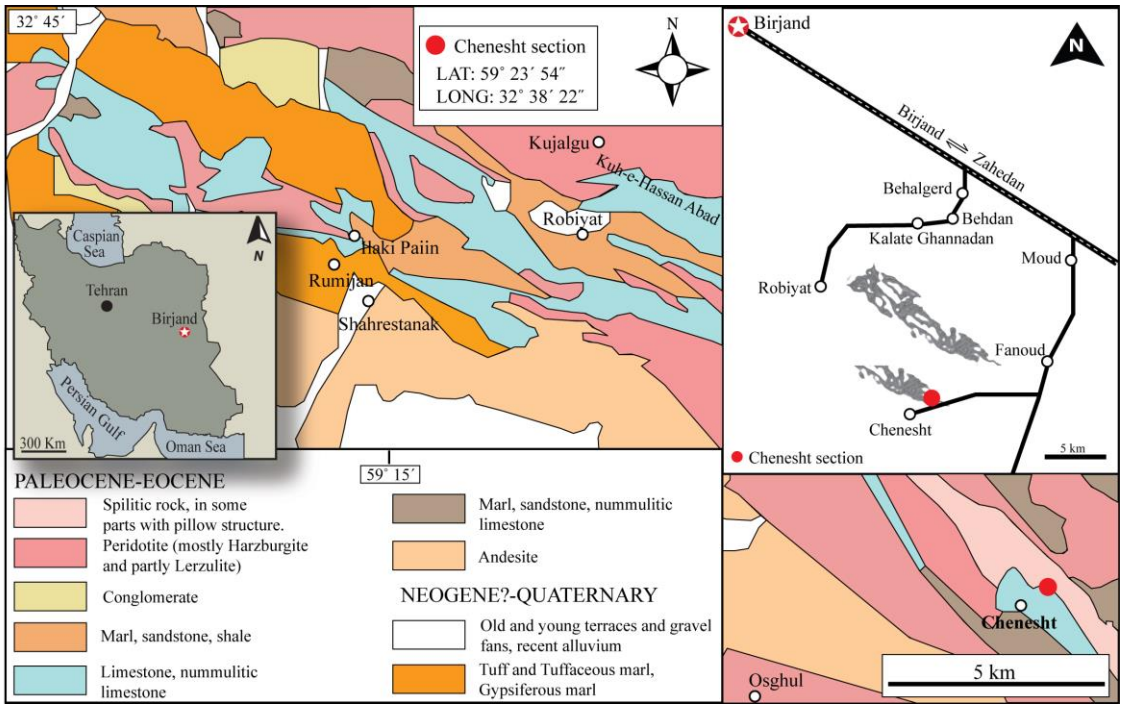


Fig. 1 Geographic map and location of the Chenesht section (Geological map of Birjand region, 1/100000 from Eftekhar-Nezhad, 1986).

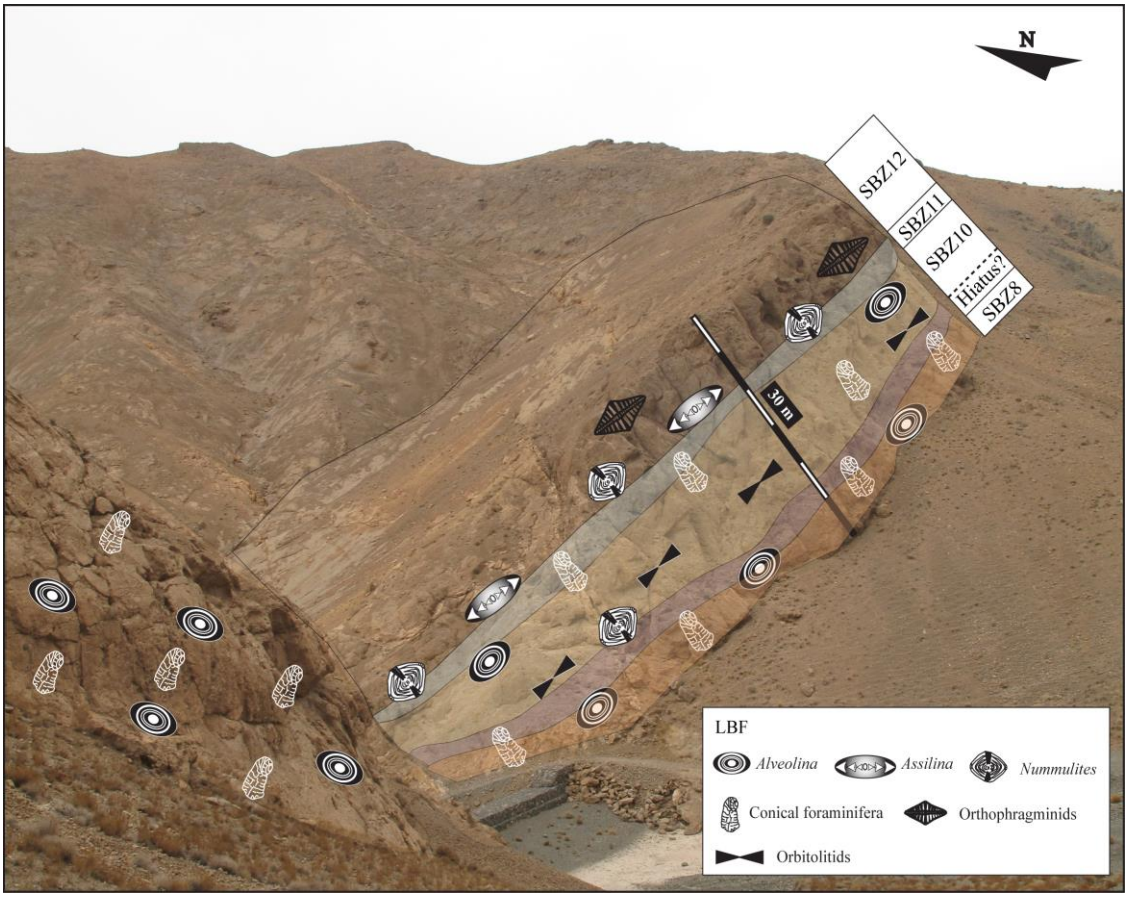


Fig. 2 Field view of the Chenesht section with schematic distribution of main groups of larger benthic foraminifera.

(Figs. 4–5, 6a, 7a–c pars)
1983 *Lituonella* cf. *roberti* Schlumberger – Rahagi, pl. 1, figs. 8–13, upper part of Lower Eocene of Iran (Sistan Suture Zone).
2017 *Coskinolina liburnica* Stache – Sirel & Deveciler, pl. 4, figs. 18–19, Cuisian of Turkey.
2017 *Coskinolina liburnica* Stache – Almansinia, pl. 10.2, fig. F–H, Eocene of Zagros Zone.
?2018 *Daviesiconus* sp. – Boudagher-Fadel, pl. 6.2, fig. 6, Early Eocene of Pakistan.

Derivation of name. The name refers to the Sistan Suture Zone, Eastern Iran.

Holotype. Subaxial section shown in Figure 4a. Thin-section CH 4.

Horizon and locality. Middle Ilerdian to late Cuisian carbonates of the Chenesht section (Fig. 2–3).

Diagnosis. Medium-sized representative of the genus (maximum diameter up to 2.0 mm, height up to 1.85 mm) with slightly convex to almost plane test base. Often the tests are as broad as high, roughly triangular with plane sides in axial sections. Rarely they display a bell-shaped outline. High conical megalospheric forms with small eccentric embryo (protoconch and deuteroconch). Microspheric forms broader than high, with convex base. Initial trochospire of up to 10 chambers. There are about 6–8 chambers per 1 mm axial length. Adult test with up to 10 uniserial comparably high chambers. The pillars display a regular pattern, are comparably widely spaced, and alternate between successive chambers. Foramina numerous (cribrate distribution). Wall with pseudokeriothecal texture.

Remarks. Most specimens being almost equally wide as high correspond to megalospheric specimens. In accordance with other species of *Coskinolina* (see Hottinger and Drobne 1980), larger low-conical forms are assumed to correspond to the microspheric generation. The pseudokeriotheca, one characteristic of the genus, is only rarely discernible in *Coskinolina sistanensis* as fine striation of the wall. As Vicedo et al. (2013) remarked, diagenetic processes often mask this structure.
Rahagi (1980, pl. 1, figs. 1–4) described *Coskinolina* n. sp. from the Middle Eocene of the Sabzewar area, Central Iran. It displays a high conical comparably large test (height up to 2.5 mm, diameter up to 3 mm), plane base and up to 16 uniserial adult chambers. With its marginal zone subdivided by short exoskeletal beams of first order the illustrated taxon however does not belong to *Coskinolina* but to *Daviesiconus* Hottinger & Drobe 1980 or *Barattolites* Vecchio & Hottinger, 2007. From the upper part of the Lower Eocene of the Sistan Zone, Rahagi (1983, pl. 1, figs. 8–13) reported *Coskinolina sistanensis* n. sp. as *Lituonella* cf. *roberti*. This taxon however a form that is clearly different (see Fig. 6d). *Coskinolina sistanensis* n. sp. has also been reported as *Coskinolina liburnica* from the Cuisian of the Hayman-Polatli Basin in Central Anatolia, Turkey by Sirel & Deveciler (2017, pl. 4, figs. 17-18) (Fig. 6a).

Comparisons. For comparison of *Coskinolina sistanensis* with other species of the genus, the most important distinct characteristics are other species are mainly due to the comparably small size of both test and embryo, and the rather coarse and regular endoskeleton. It is worth mentioning that others like *Coskinolina floridana* Cole, 1941 belong to different genera such as *Coskinon* Hottinger & Drobne, 1980 or *Fallotella* Mangin (Bowen Powell, 2010), among others because of the subdivided marginal zone. *Coskinolina elongata* Cole (Late early Eocene of Jamaica), 1941, has been designated the type-species of *Coleiconus* Hottinger & Drobne, 1980.

Coskinolina liburnica Stache, 1875 (Cuisian of Croatia) (Fig. 6b-c): Medium- to large-sized test with distinctly convex base. Marginal chamber cavity narrow, slightly compressed or isometric. Megalosphere up to 0.4 mm. B-forms with more than 20 juvenile chambers. Chambers low in height (6–7 per mm axial length). Pillars densely arranged. There are 6–7 chambers per 1 mm axial length. Reference: Stache (1875), Schubert (1912), Hottinger & Drobne (1980), Hottinger (2007), Serra Kiel et al. (2016). Stratigraphy: SBZ 11–SBZ 12 (Serra Kiel et al., 1998: Late Ypresian = Middle–Late Cuisian); Late Ypresian to Priabonian, SBZ 11–SBZ 20 (Serra Kiel et al., 2016). *C. sistanensis* has been confounded with *C. liburnica* in the literature (see synonymy).

Coskinolina roberti (Schlumberger, 1905; Eocene of France) (Fig. 6d): High-conical, large test (diameter up to > 3 mm) with large initial trochospire (see Poignant 1961; Hottinger and Drobne 1980). This species has reliably been recorded only from high-energy deposits (e.g. rudstones) of western France. Proloculus large (diameter 0.35–0.4 mm). There are 5–6 chambers per 1 mm axial length.

Coskinolina perpera Hottinger & Drobne, 1980 (Eocene of Croatia) (Fig. 6e): Chambers comparably high giving rise to often irregularly twisted pillars. The pillars display a general loose and irregular pattern. Marginal chamber cavities are distinctly inflated. There are 4 to 5 chambers per mm axial length (2 to 3 in microspheric forms). Embryo large consisting of a protoconch (diameter 0.35–0.45 mm) and deuteroconch (> 0.5 mm). Occurrences: Late Cuisian to Early Lutetian of Croatia (Hottinger and Drobne, 1980); Middle Eocene of Somalia (Azzaroli 1950 fide Hottinger, 2007); Bartonian of Iran (Hottinger, 2007); Lutetian-Priabonian of Oman (Serra Kiel et al., 2016).

Coskinolina douvillei (Davies, 1930; Cuisian of Pakistan) (Fig. 6f): Medium-sized cone, slightly higher than wide (base mean 2.19 mm, height mean 2.64 mm, Davies 1930). Microspheric specimens attain a size of more than 7 mm (Hottinger and Drobne 1980, pl. 10, fig. 2). Diameter protoconch 0.3–0.35 mm. Adult specimen with more than 15 uniserial chambers (e.g., Davies 1930, pl. 2, fig. 3). There are 5–6 chambers per 1 mm axial length.

Coskinolina depressa (Azzaroli, 1952; Lutetian of Somalia): Large-sized (cone diameter up to 3.5 mm, cone height up to 3 mm, A-forms), low-conical species of the

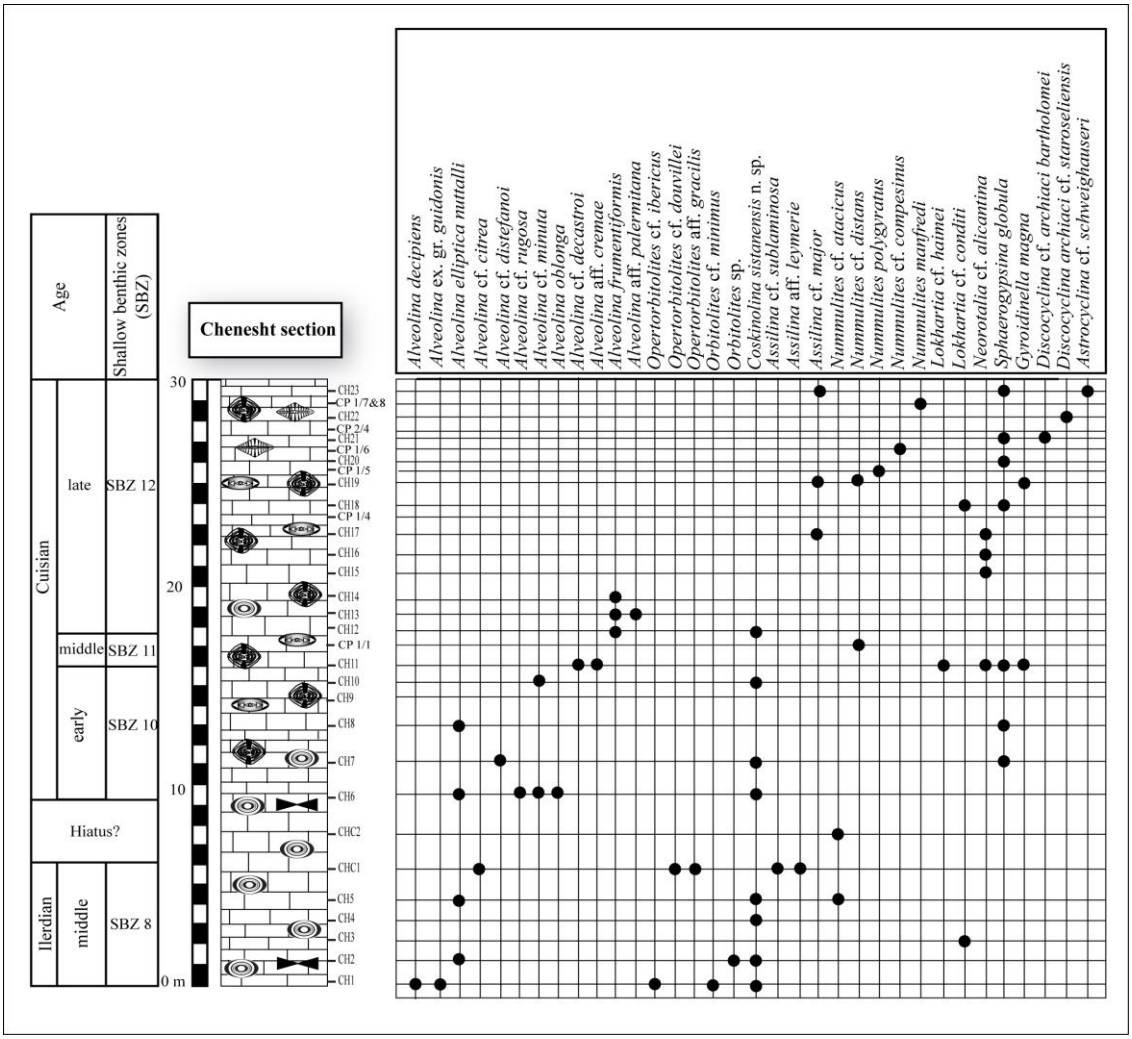


Fig. 3 Log of the Chenesht section with distribution of larger benthic foraminifera.

genus, often displaying convex base. Occurrences: Small sized *Coskinolina* with distinctly convex base were described by Serra-Kiel et al. (2016) as *C.* cf. *depressa* from the uppermost Bartonian to the Priabonian (SBZ 18–SBZ 20) of Oman. *C. depressa* is very close to *C. douvillei* if not a junior synonym (Hottinger and Drobne, 1980, p. 44).

Stratigraphic distribution. In the Chenesht section *Coskinolina sistanensis* n. sp. occurs from the middle Illerdian to the early late Cuisian (= SBZ 8 to SBZ 12, basal part, of Serra Kiel et al., 1988). The occurrence of *C. sistanensis* n. sp. in older strata however cannot be excluded (see Fig. 2).

Microfacies, association, and palaeoenvironment: *Coskinolina sistanensis* n. sp. occurs in *Alveolina-Opertorbitolites* packstone and nummulitids-*Alveolina* rudstone, microfacies types restricted to the basal to middle parts of the Chenesht section (Fig. 7). The first microfacies type is characterized by an abundance of *Alveolina decipiens* Schwager, *A.* ex gr. *guidonis* Drobne, *A. elliptica nuttalli* Davis, *A.* cf. *citrea* Drobne, with globular to slightly elongated forms and diameters less than 5 mm, *Opertorbitolites* (e.g., *O. douvillei* Nuttall, *O.* cf. *ibericus* Lehmann), *Orbitolites* (e.g., *Orbitolites* sp. cf. *O. minimus* Henson). Besides larger benthic foraminifera, small miliolids (*Quinqueloculina*, *Triloculina*, *Biloculina*), small rotaliids (*Neorotalia* sp.) occur. Detrital quartz grains are present in varying amounts with fine to medium sizes. The nummulitids-*Alveolina* rudstone microfacies is distinguished by small nummulitids i.e. *Nummulites* cf. *atacicus* Leymerie, *Assilina sublaminosa* Gill, *N. tauricus* de la Harpe, *N. pratti* d'Archiac & Haime, *N.* cf. *distans* Deshayes associated with *Alveolina* (e.g., *A.* cf. *minuta* Checchia-Rispoli, *A. elliptica nuttalli* Davies, *A.* cf. *decastroi* Scotto Di Carlo) together with a decrease of abundance of *Coskinolina sistanensis* n. sp. Other subordinate components are encrusting foraminifera (acervulinids), orthophragminids, miliolids, orbitolitids, small tests of undetermined rotaliids, green algae, echinoids and quartz grains. With a predominance of larger porcelaneous taxa associated with agglutinating conical forms this microfacies type can be assigned to a proximal inner ramp setting (e.g., Vecchio and Hottinger, 2007). With the occurrence of alveolinids and conical agglutinating taxa in the lower two third of the section and the dominance of nummulitids in the last third, a deepening upward trend is documented (Vecchio and Hottinger, 2007; Zamagni et al., 2008) (Figs. 2–3).

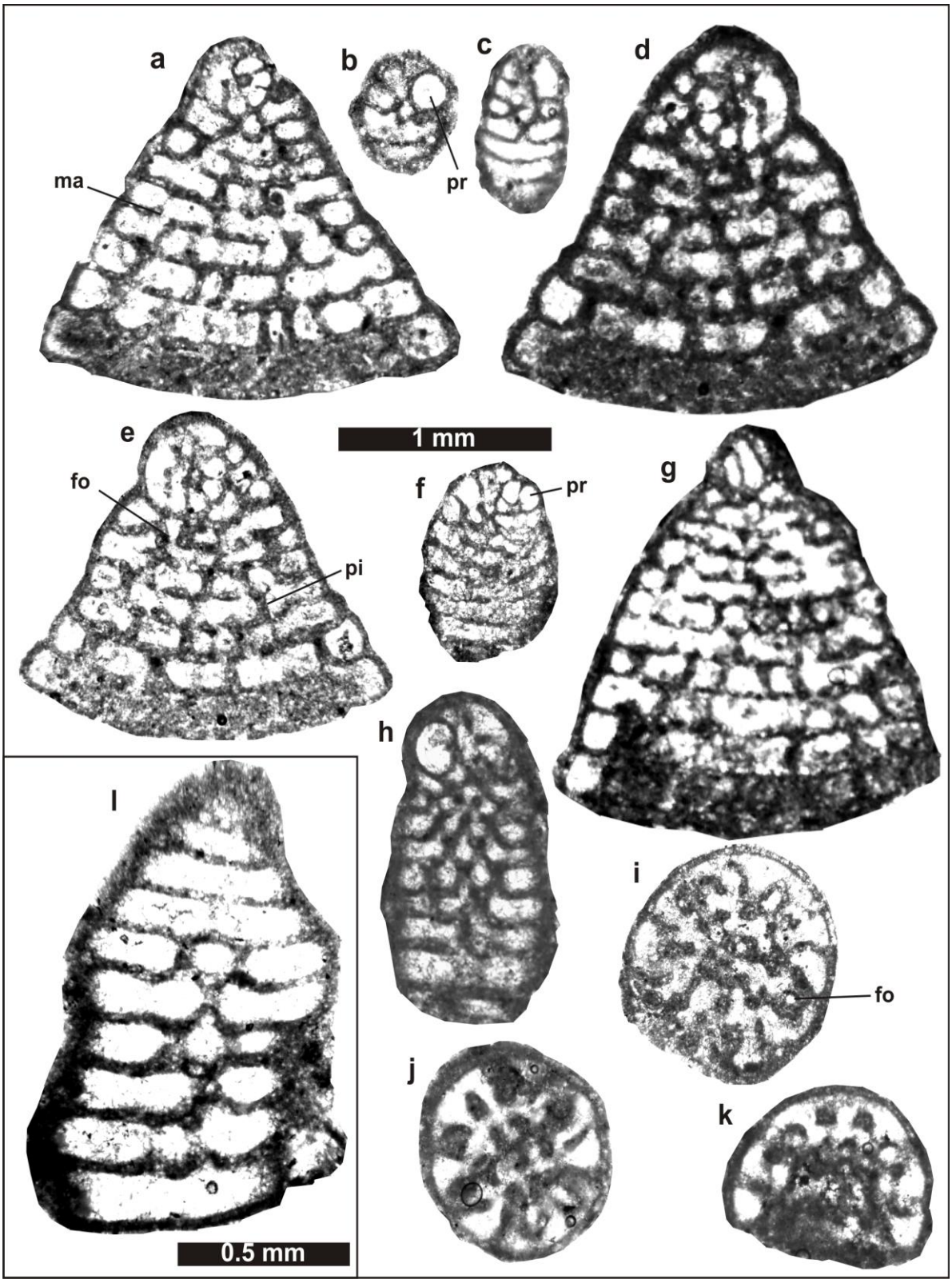


Fig. 4 *Coskinolina sistanensis* n. sp., Illerdian-Cuisian of Iran: **a, d–e, g** (sub)axial sections showing small initial spire. Holotype specimen in **a**. **b, h** oblique sections of megalospheric form. **c, f** tangential-oblique sections. **i–k** transverse slightly-oblique sections. **l** tangential section showing undivided marginal zone and pseudokeriotheka (right side). Thin-sections: CH 4 **(a–c, e)**, CH 12 **(d, f, j–l)**, CH 2 **(g–h)**, CH 7 **(i)**. Abbreviations: fo = foramen, ma = marginal aperture, pr = proloculus, pi = pillar.

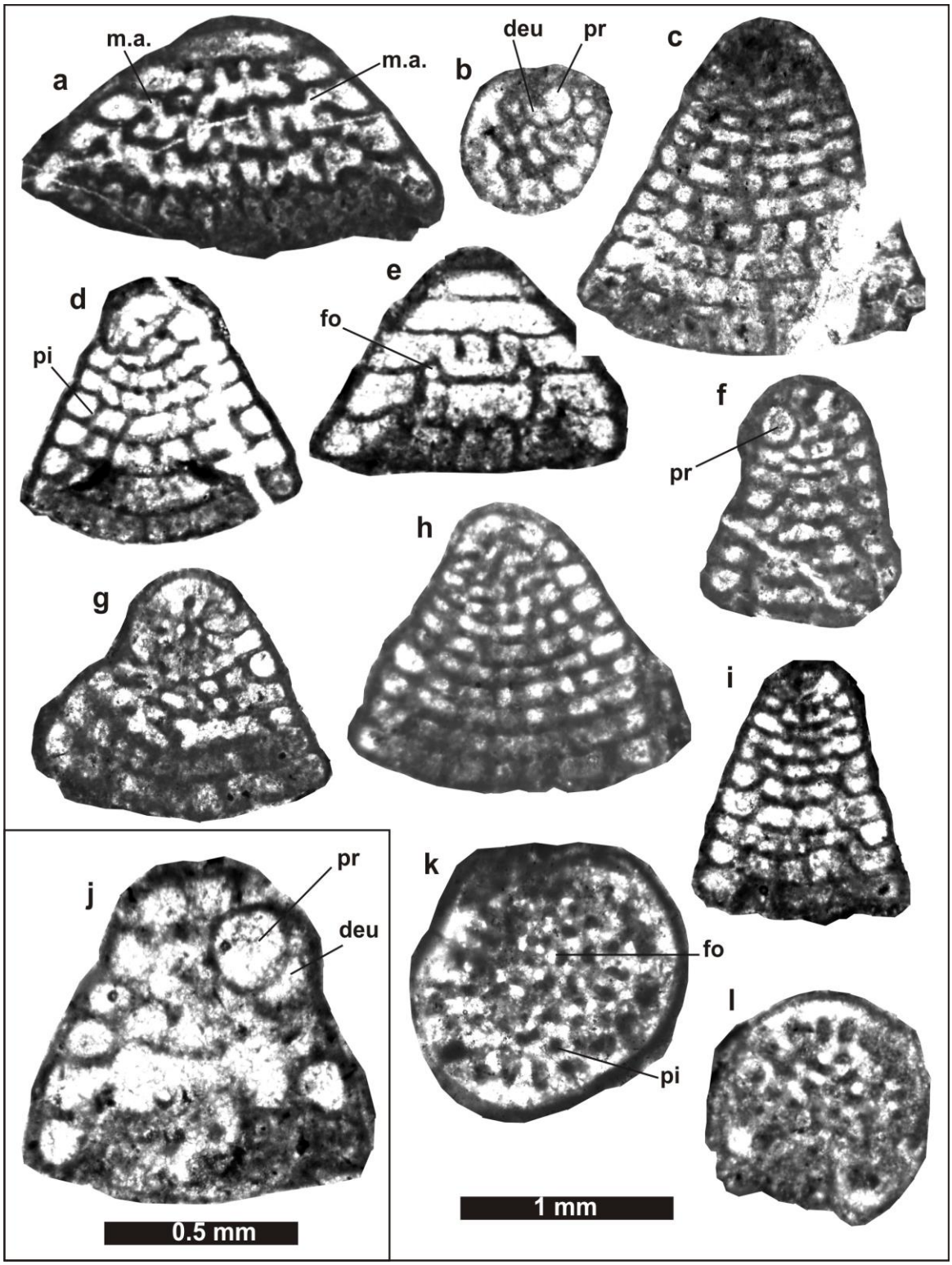


Fig. 5 *Coskinolina sistanensis* n. sp., Illerdian-Cuisian of Iran: **a** oblique section of a low-conical specimen (?microspheric form). **b**, oblique section showing embryo. **c–d, g–h**, subaxial sections. **e**, tangential-oblique section. **f, i,** slightly oblique axial sections; embryo shown in (**i**). **j** axial section of a small megalospheric specimen showing embryo. **k–l** slightly oblique transverse sections. Thin-sections: CH 2 **(a, c–h)**, CH 7 **(b),** CH 12 **(i, k–l)**, CH 4 **(j)**. Abbreviations: deu = deuteroconch, fo = foramen, m.a. = marginal aperture, pi = pillar, pr = proloculus.

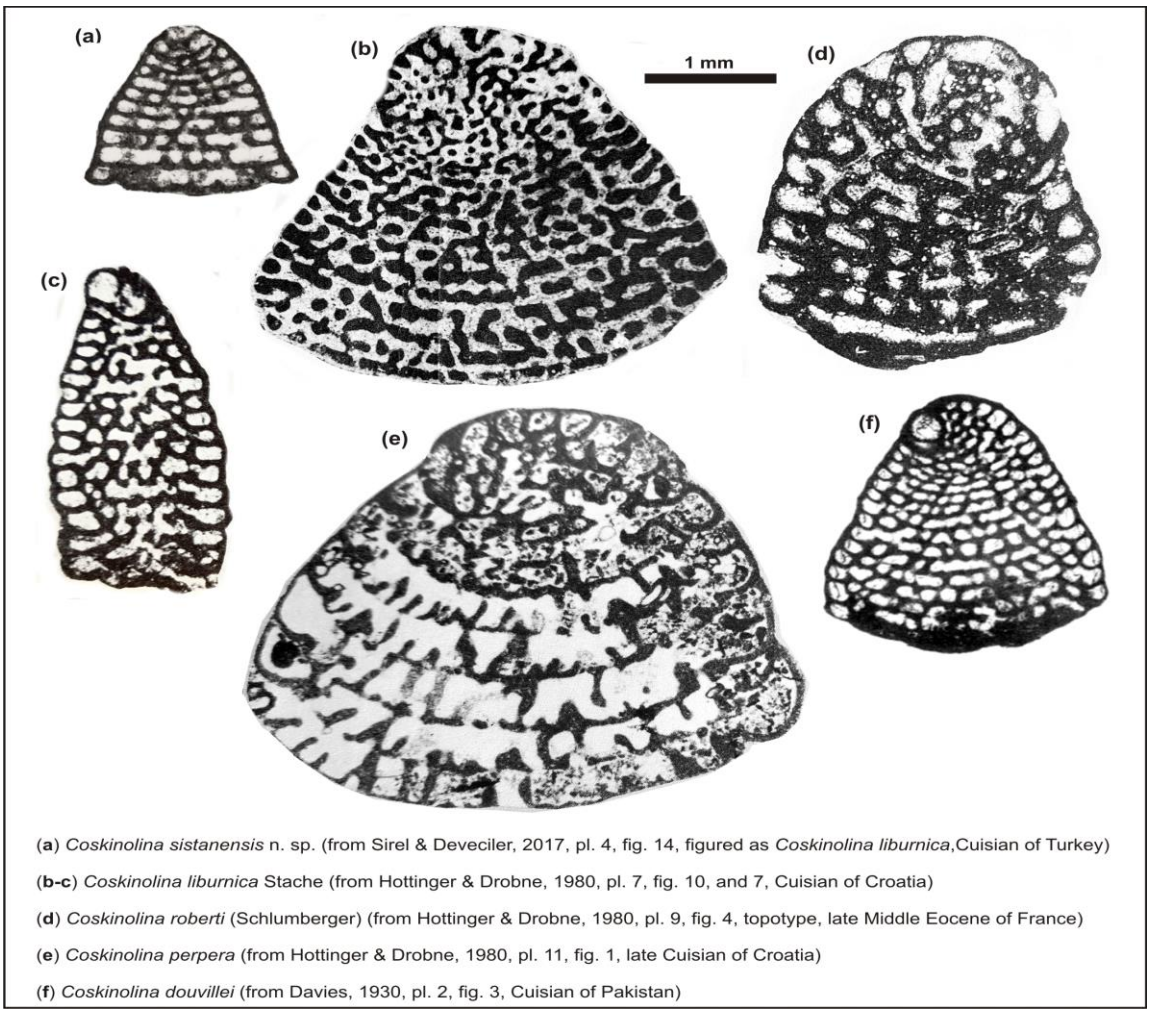


Fig. 6 **a** *Coskinolina sistanensis* n. sp., compared to the type-species *Coskinolina liburnica* Stache in **b–c**, **d** *Coskinolina roberti* Schlumberger, **e** *Coskinolina perpera* Hottinger & Drobne, **f** *Coskinolina douvillei* (Davies).

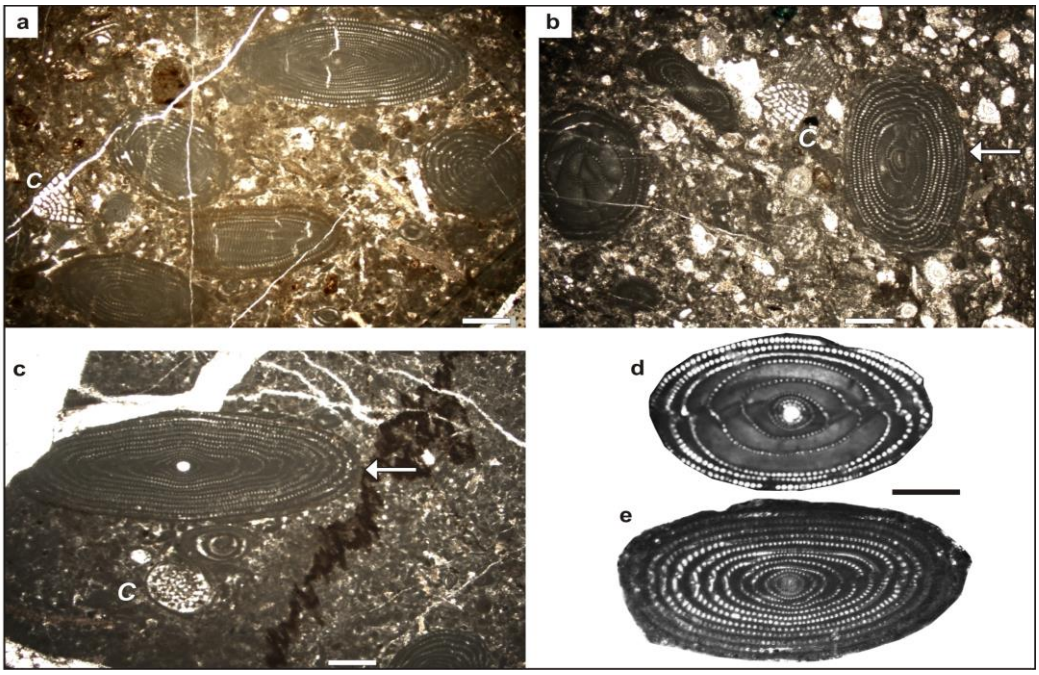


Fig. 7 Microfacies and selected alveolinids from the Chenesht section. **a–c** *Alveolina-Opertorbitolites* packstone with *Coskinolina sistanensis* n. sp. (*C*), *Alveolina elliptica* (Sowerby) (arrows in **b**), *Alveolina frumentiformis* Schwager (arrow in c). **d** *Alveolina elliptica nuttali* Davies. **e** *Alveolina oblonga* d'Orbigny. Samples CH 2 (**a**), CH 6 (**b, e**), CH 12 (**c**), CH 5 (**d**). Scale bars 1 mm.

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REFERENCES

Aghanabati, A., 2004. Geology of Iran. Geological Survey of Iran.

Alavi, M. 1991. Sedimentary and structural characteristics of the Paleo-Tethys remnants in northeastern Iran. Geological Society of America Bulletin, 103: 983-992.

Almasinia, B., 2017. Microfacies, biostratigraphy and sedimentary environments of the Eocene sedimentary successions in the Zagros and Sanandaj-Sirjan Zone (Iran). Phd Thesis University of Erlangen. https://opus4.kobv.de/opus4-fau/files/8675/Library+version.pdf

Alavi, M., 1993. Paleozoic stratigraphy of Iran. Treatise on the Geology of Iran, 5: 1-492.

Azzaroli, A., 1952. I macroforaminiferi della serie del Carcàr (Eocene medio e superior) in Somalia e la loro distribuzione stratigrafica. Palaeontographia Italiana, 17: 99-131.

Babazadeh, S.A., De Wever, P., 2004. Radiolarian Cretaceous age of Soulabest radiolarites in ophiolite suite of eastern Iran. Bulletin de la Société Geologique de France, 175 (2): 121-129.

Berberian, M., 1977. Against the rigidity of the "Lut Block". Geological Mineral Research Organization report, 40: 203-227.

Boudagher-Fadel, M., 2018. Evolution and geological significance of larger benthic foraminifera (second edition). UCL Press, London, 702 p.

Bowen Powell, J., 2010. Larger foraminiferal biostratigraphy, systematics, and paleoenvironments of the Avon Park Formation and Ocala Limestone, Highlands County, Florida. FIU Electronic Theses and Dissertations. 288. http://digitalcommons.fiu.edu/etd/288

Bröcker, M., Rad, G.F., Burgess, R., Theunissen, S., Paderin, I., Rodionov, N. & Salimi, Z., 2013. New age constraints for the geodynamic evolution of the Sistan Suture Zone, eastern Iran. Lithos, 170: 17-34.

Cole, W.S., 1941. Stratigraphic and paleontologic studies of wells in Florida. Florida Geological Survey Bulletin, 19: 91 p.

Davies, L.M., 1930. *Dictyoconus* and its allies: a review of the group together with a description of three new species from the lower Eocene beds of Northern Baluchistan. Trans. Royal Soc. Edinburgh, 56 (2): 485-505.

Douglass, R.C., 1960. Revision of the family Orbitolinidae. Micropalaeontology, 6 (3): 249-270.

Eftekharnezhad, J., 1980. Subdivision of Iran into different structural realms with relation to sedimentary basins (in Farsi). Bulletin of the Iranian Petroleum Institute, 82: 19-28.

Eftekhar-Nezhad, J., 1986. Geological map of Iran. 1:100000 series, Sheet n. 7855, Birjand, Tehran, Geological and Mineral Survey of Iran.

Ghorbani, M., 2013. A summary of geology of Iran. In: The Economic Geology of Iran. Springer, pp. 45–64.

Hadi, M., Mosaddegh, H., Abbassi, N., 2015. Biostratigraphic interpretation and systematics of Alveolina assemblages in the Ziarat Formation from Soltanieh Mountains (Western Alborz). Geoscience Scientific Quarterly Journal, 24 (95): 39-44.

Hadi, M., Mosaddegh, H., Abbassi, N., 2016. Microfacies and biofabric of nummulite accumulations (Bank) from the Eocene deposits of Western Alborz (NW Iran). Journal of African Earth Sciences, 124: 216-233.

Hottinger, L., 2007. Revision of the family genus *Globoreticulina* Rahagi, 1978, and of its associated fauna of larger foraminifera from the late Middle Eocene of Iran. Carnets de Géologie / Notebooks on Geology, article 2007/06 (CG2007_A06), 1-51.

Hottinger, L., Drobne, K., 1980. Early tertiary conical imperforate foraminifera. Razprave IV razr. SAZU, 22: 188-276.

Kaminski, M.A. 2000. The new and reinstated genera of agglutinated foraminifera published between 1986 and 1996. In: Hart, M.B., Kaminski, M.A. & Smart, C.W. (eds) Proceedings of the Fifth International workshop on agglutinated foraminifera. Grzybowski Foundation Special Publication, 7: 185-219.

Kaminski, M.A., 2014. The year 2010 classification of the agglutinated Foraminifera. Micropaleontology, 60: 89-108.

Mosaddegh, H., Hadi, M., Parandavar, M., 2017. Biostratigraphy of the Eocene carbonate deposits in the eastern part of the central Iran (Torbate-e-jam area): stratigraphic significance of *Nummulites perforatus* and Calcareous nannofossils. Scientific Quarterly Journal Geosciences, 102: 91-100.

Pawlowski, J., Holzmann, M., Tyszka, J., 2013. New supraordinal classification of Foraminifera: Molecules meet morphology. Marine Micropaleontology, 100: 1-10.

Poignant, A. 1960. Aperçu sur la microfaune de l' Éocène supérieur du Médoc. Revue de Micropaléontology, 3 (1), 31–36.

Rahaghi, A., 1978. Paleogene biostratigraphy of some parts of Iran. Publications of the National Iranian Oil Company Geological Laboratories, 17: 1-161.

Rahaghi, A., 1980. Tertiary faunal assemblage of Qum-Kashan, Sabzewar and Jahrum areas. Publications of the National Iranian Oil Company Geological Laboratories, 8: 1-64.

Rahaghi, A., 1983. Stratigraphy and faunal assemblage of Paleocene-Lower Eocene in Iran. Publications of the National Iranian Oil Company Geological Laboratories, 10: 1-73.

Rahaghi, A., Schaub, H., 1976. Nummulites et Assilines du NE de l'Iran. Eclogae Geologicae Helvetiae, 69: 765-782.

Schlumberger, C., Douvillé, H., 1905. Sur deux Foraminifères Eocènes. *Dictyoconus egyptiensis* Chapm. et *Lituonella roberti* nov. gen. et sp. Bulletin de la Société Géologique du France, 5 (4): 291-304.

Schubert, R., 1912. Über *Lituonella* und *Coskinolina liburnica* Stache sowie deren Beziehungen zu den

anderen Dictyoconinen. Jahrbuch der kaiserlich-königlichen geologischen Reichsanstalt, 62: 195-208.

Serra-Kiel, J., Gallardo-Garcia, A., Razin, Ph., Robinet, J., Roger, J., Grelaud, J., Leroy, S., Robin, C., 2016. Middle Eocene-Early Miocene larger foraminifera from Dhofar (Oman) and Socotra Island (Yemen). Arabian Journal of Geoscience, 9 (5): 344.

Serra-Kiel, J., Hottinger, L., Caus, E., Drobne, K., Ferrandez, C., Jauhri, A.K., Less, G., Pavlovec, R., Pignatti, J., Samso, J.M. 1998. Larger foraminiferal biostratigraphy of the Tethyan Paleocene and Eocene. Bulletin de la Société Geologique de France, 169: 281-299.

Sirel, E., Deveciler, A., 2017. A new late Ypresian species of *Asterigerina* and the first records of *Granorotalia* from the Thanetian and Upper Ypresian of Turkey. Rivista Italiana di Paleontologia e Stratigrafia, 123 (1): 65-78.

Stache, G., 1875. Neue Beobachtungen in den Schichten der liburnischen Stufe. Verhandlungen der Geologischen Reichsanstalt, 1875, 334–338.

Tirrul, R., Bell, I., Griffis, R., Camp, V., 1983. The Sistan suture zone of eastern Iran. Geological Society of America Bulletin, 94, 134–150.

Vecchio, E., Hottinger, L., 2007. Agglutinated conical foraminifera from the Lower-Middle Eocene of the Trentinara Formation (southern Italy). Facies, 53, 509-533.

Vicedo, V., Berlanga, J.A., Serra-Kiel., J., 2014. Paleocene larger foraminifera from the Yucatán Peninsula (SE Mexico). Carnets de Géologie [Notebooks on Geology], 14 (4): 41-68.

Zamagni, J., Mutti, M., Kosir, A., 2008. Evolution of shallow benthic communities during the late Paleoecene-earliest Eocene transition in the Northern Tethys (SW Slovenia). Facies, 54 (1): 25-43.